

BTS-VI-04.19-0611

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B.Tech. Degree VI Semester Examination April 2019

CS 15-1603 CRYPTOGRAPHY AND NETWORK SECURITY
(2015 Scheme)



Time: 3 Hours

Maximum Marks: 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Describe the significance of the terms confidentiality, integrity and availability in computer security.
(b) Describe the Shannon's characteristics of a good cipher.
(c) Describe how DES exhibit avalanche effect.
(d) Discuss the strengths of DES algorithm.
(e) Compare AES and DES algorithm.
(f) Explain how public key encryption can be used to create digital signature.
(g) Describe primality testing.
(h) Explain Diffie-Hellman cryptosystem.
(i) Write a short note on SSL.
(j) Explain any two authentication applications.

PART B

(4 × 10 = 40)

- II. Explain any two substitution and any two transposition ciphers with examples. (10)
- OR**
- III. (a) Comment on modular arithmetic and its properties. (7)
(b) State Fermat's theorem. (3)
- IV. Define symmetric key encryption. Explain any one symmetric key encryption algorithm. (10)
- OR**
- V. Explain the operations of AES algorithm. (10)
- VI. Describe RSA algorithm with an example. (10)
- OR**
- VII. Explain Miller-Rabin algorithm with a suitable example. (10)
- VIII. Describe the IP security architecture in detail. (10)
- OR**
- IX. (a) Describe various types of firewalls and their characteristics in detail. (6)
(b) What is meant by digital signatures? (4)
